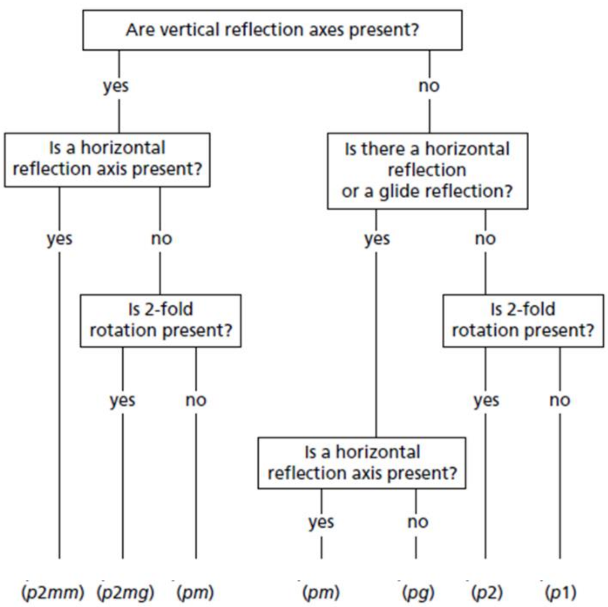
Equations, constants and conversions for Exam 2

Constants:

N_A = 6.02 x 10^{23} atoms/mole; $h = 6.626\times10^{-34}$ J·s; 1 J = 6.24×10^{18} eV; 1 eV = 1.602×10^{-19} J

e = 2.71828; π = 3.14159

Line group flow chart:



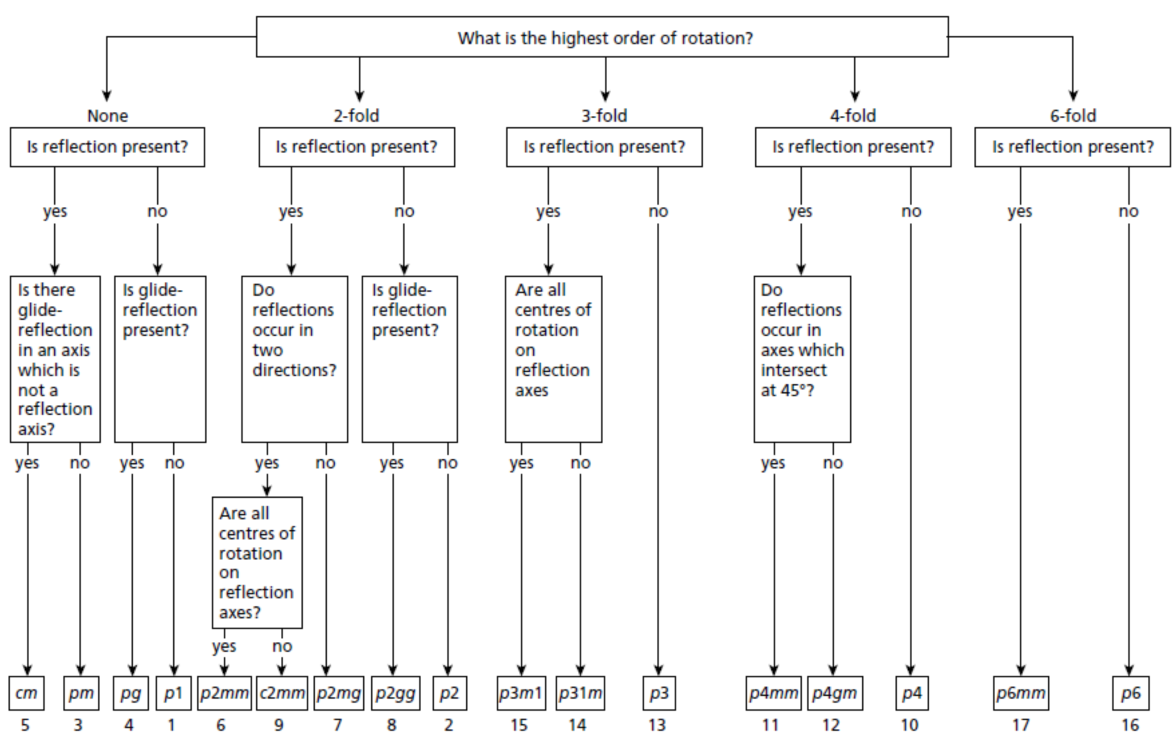
Rotation Matrices:

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

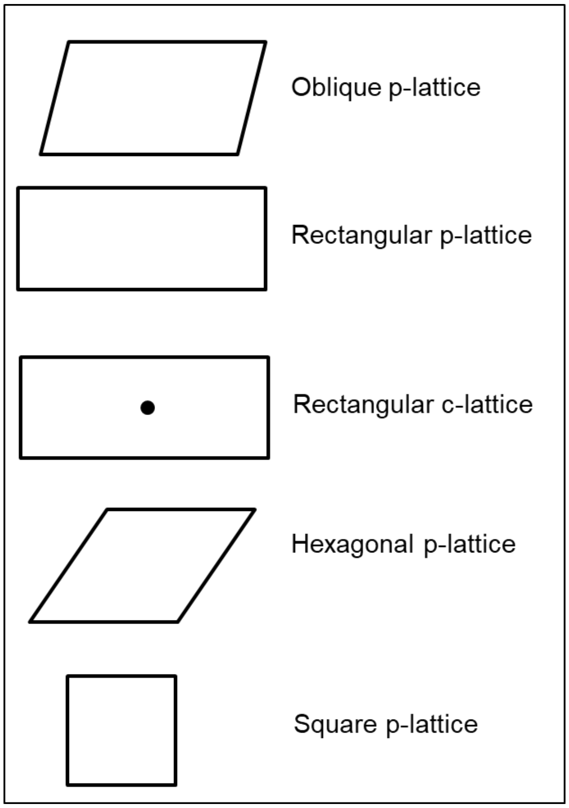
$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

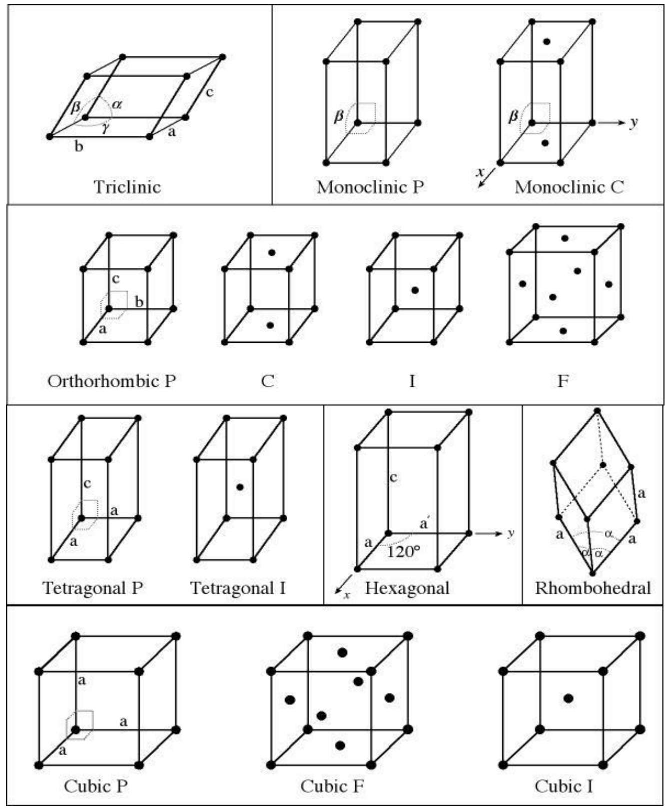
Plane group flow chart:



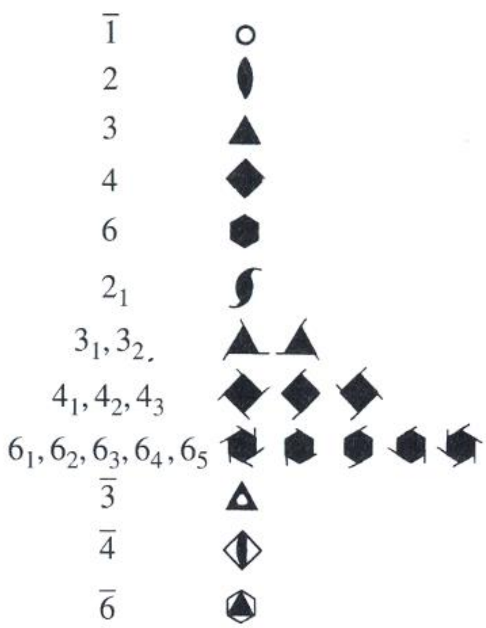
2D Bravais lattices:

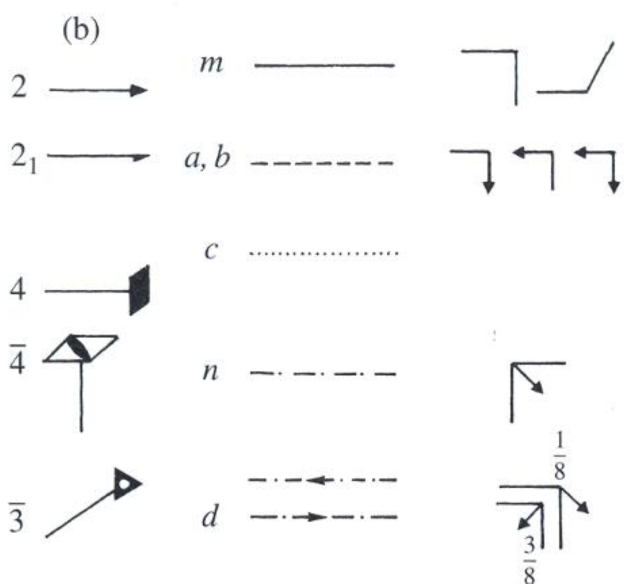


3D Bravais lattices:



Symmetry elements:

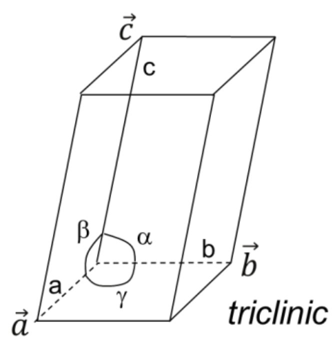




Symmetry directions in point groups by crystal system:

Lattice	Primary	Secondary	Tertiary
Triclinic	None		
Monoclinic	unique (*b* or *c*)		
Orthorhombic	**[100]**	**[010]**	**[001]**
Tetragonal	**[001]**	{**[100]**, **[010]**}	{$[\mathbf{1\bar{1}0}]$, **[110]** }
Hexagonal	**[001]**	{**[100]**,**[010]**,$[\mathbf{\bar{1}\bar{1}0}]$}	{$[\mathbf{1\bar{1}0}]$,**[120]**,$[\mathbf{2\bar{1}0}]$}
Rhom. (hex.axis)	**[001]**	{**[100]**,**[010]**,$[\mathbf{\bar{1}\bar{1}0}]$}	
Rhom. (rho)	**[111]**	{$[\mathbf{1\bar{1}0}]$,$[\mathbf{01\bar{1}}]$,$[\mathbf{\bar{1}01}]$}	
Cubic	{**[100]**, **[010]**,**[001]**}	{**[111]**, $[\mathbf{1\bar{1}\bar{1}}]$, $[\mathbf{\bar{1}1\bar{1}}]$,$[\mathbf{\bar{1}\bar{1}1}]$}	{$[\mathbf{1\bar{1}0}]$,**[110]**,$[\mathbf{01\bar{1}}]$, **[011]**,$[\mathbf{\bar{1}01}]$,**[101]**}

Metric tensor:



$$\Gamma \equiv \begin{bmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c} \end{bmatrix}$$

$$\Gamma = \begin{bmatrix} a^2 & ab\cos\gamma & ac\cos\beta \\ ab\cos\gamma & b^2 & bc\cos\alpha \\ ac\cos\beta & cb\cos\alpha & c^2 \end{bmatrix}$$

$$\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta \quad \vec{a}\cdot\vec{b} = [a_1 \quad a_2 \quad a_3][\Gamma]\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad |\vec{a}| = \sqrt{\vec{a}\cdot\vec{a}} = \sqrt{[a_1 \quad a_2 \quad a_3][\Gamma]\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}}$$

$$L^2 = [\Delta x \;\; \Delta y \;\; \Delta z][\Gamma]\begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix}$$

$$V = \vec{a}\bullet(\vec{b}\times\vec{c}) = abc\sqrt{1-\cos^2\alpha-\cos^2\beta-\cos^2\gamma+2\cos\alpha\cos\beta\cos\gamma}$$